



Australian
National
University

Venue

Student Number

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Research School of Computer Science

EXAMINATION

Semester 1 - In-Class & Online, 2026

COMP2300/COMP6300/ENGN2219_Semester 1 Computer Architecture

This paper is for ANU, ANU and ANU students.

Examination Duration: 90 minutes

Reading Time: 0 minutes

Exam Conditions:

120 mins

Materials Permitted In The Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

No materials permitted

Materials To Be Supplied To Students:

Scribble Paper x 2

Instructions To Students: The exam is marked out of 100.

Question 1: Multiple choice and true/false questions.

a. The number of bits needed to represent -32 in 2's complement number system:

- A. 7
- B. 8
- C. 9
- D. 10
- E. None of the above

b. A logic gate with two inputs, A and B, outputs B when A is Logic 1, and the complement of B, when A is Logic 0. Which logic gate is it?

- A. OR
- B. AND
- C. XOR
- D. XNOR
- E. NAND

c. The expression $A + B$ is equivalent to:

- A. $A' \cdot B'$
- B. $(A \cdot B)'$
- C. $(A' \cdot B')'$
- D. $((A \cdot B)')'$
- E. None of the above

d. The expression $ABC + AB'C + ABC'$ is equivalent to:

- A. $ABC + BC$
- B. $A(B + C)$
- C. ABC
- D. $AB + BC$
- E. $ABC + B'C$

e. Which of the following correctly describes the behavior of SR latches built using cross-coupled NOR and NAND gates, when both inputs are set to Logic 1?

- A. In both NOR and NAND latches, the outputs retain their previous state
- B. In a NOR latch, both outputs go to Logic 0; in a NAND latch, both outputs go to Logic 1
- C. A NOR latch enters an invalid state; a NAND latch holds the previous state
- D. Both types of latches enter a valid and predictable state
- E. In a NOR latch, the outputs are complementary; in a NAND one, both outputs are Logic 0

f. A memory array that uses a 4:16 decoder to store 2-byte words has an address space (in bits) of:

- A. 16
- B. 32
- C. 64
- D. 128
- E. 256

g. The width of pipeline registers in a pipelined CPU is determined by:

- A. ISA
- B. Microarchitecture
- C. None of the above

h. The decision to increment the PC by a specific offset to fetch the next instruction is determined by:

- A. ISA
- B. Microarchitecture
- C. None of the above

i. If one of the inputs of a NAND gate is connected to logic level 1 (TRUE) forever, the resulting circuit behaves like:

- OR gate
- NOR gate
- NOT gate
- Short circuit wire
- Tri-state buffer

j. Under De Morgan's Theorem, which of the following circuits are equivalent:

- An AND gate with inverted output *is equivalent to* an AND gate with inverted inputs
- An AND gate with inverted output *is equivalent to* a NOR gate with inverted output
- An AND gate with inverted output *is equivalent to* an OR gate with inverted inputs
- An OR gate with inverted output *is equivalent to* an OR gate with inverted inputs
- A NOR gate with inverted output *is equivalent to* an AND gate with inverted inputs
- None of the above

k. A 2:4 decoder can be used to implement an XNOR gate by connecting which ones of its one-hot outputs to the inputs of an OR gate:

- 00 and 01
- 00 and 11
- 01 and 10
- 10 and 11
- 00 and 10

l. Two NAND gates are cross-coupled to produce outputs X and Y. Their remaining terminals are connected to inputs A and B. When $A = 1$ and $B = 1$, which one of the following statements is correct?

- X is 1, and Y is 0
- X is 0, and Y is 1
- X and Y remember their old values
- X and Y oscillate between 1 and 0
- X and Y are not complements of each other

m. What is the execution time (in seconds) of a program that executes 2 billion instructions on a CPU with a clock frequency of 2 GHz and an average IPC of 0.25?

- 0.5
- 1
- 2
- 4
- 8

Question 2. Combinational Logic

One day at work, you are tasked with designing a combinational circuit with a three-bit input, $\{A, B, C\}$ (A is the most significant bit, and C is the least significant bit), and one output, Y . The value of Y is 1 when the 3-bit input contains two or more ones, with one exception. The exceptional case occurs when A and C are 1 and B is 0. In that case, the output Y is 0.

Answer the next three questions. You can also use the white space on the next page if required.

- A. Draw the truth table for the combinational circuit.
- B. Express the output Y as the simplest sum of product representation. Show your work step-by-step. Your work should first show the expanded Boolean equations translated from the truth table.
- C. Using basic logic gates, draw a schematic of Y as a function of A , B , and C .

Question 3. Sequential Circuits

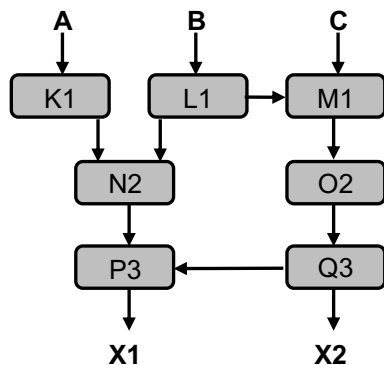
Design a finite state machine (FSM) with one input, A, and one output, X. The output X should be 1 if A has been 1 for three consecutive clock cycles. Otherwise, X should be 0.

- A. Design the Moore state transition diagram for the above specification.
- B. Build the encoded state transition tables.
- C. Specify and simplify the next state and output equations.
- D. Draw the high-level schematic of the FSM. You do not need to show the gate-level logic, but you should show the state register, next stage component, feedback loop, and output component.

Write your answers on this page and the next page.

Question 4. Timing Issues

Consider the high-level schematic of a general-purpose combinational hardware circuit shown that computes two outputs from three inputs. Circle the correct answer.



- A. The correct statement about the relationship of X1 and X2 to the three inputs is:
- a. X1 and X2 both depend on all three inputs
 - b. X1 and X2 both depend on only two of the three inputs
 - c. X1 depends on all three inputs, A, B, and C, but X2 depends on only two of the three inputs
 - d. X2 depends on all three inputs, A, B, and C, but X1 depends on only two of the three inputs
 - e. None of the above is correct
- B. If all components have a delay of 4 units, then the critical path of X1 consists of:
- a. K1, N2, P3
 - b. L1, M1, O2, Q3, P3
 - c. L1, N2, P3
 - d. L1, N2, Q3, P3
 - e. M1, O2, Q3, P3
- C. If K1 has a delay of 16 units, and all other components have a delay of 4 units, then the critical path of X1 consists of:
- a. K1, N2, P3
 - b. L1, M1, O2, Q3, P3
 - c. L1, N2, P3
 - d. L1, N2, Q3, P3
 - e. M1, O2, Q3, P3